# PAIDIPILLI PURUSHOTHAM

**Embedded Firmware & Hardware Engineer: Microcontrollers, RTOS, & C Programming**

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## EDUCATION

**B.Tech in Electronics and Communication Engineering (ECE)** — Aug 2023 – Aug 2027
*Vishnu Institute of Technology (CGPA: 9.73/10 till 6th sem)* — *Bhimavaram, AP*

**Intermediate (MPC)** – Aditya Junior College (Marks: 971/1000) — *Jun 2021 – Apr 2023*

## TECHNICAL SKILLS

- **Programming Languages:** Embedded C, C Programming, C++, Python, HTML/CSS
- **Microcontrollers & Hardware:** ESP32, Arduino Uno (ATmega328P), Zynq-7000, ARM Cortex architecture familiarity, FPGA
- **Protocols & Peripherals:** UART, SPI, I2C, ADC, DAC, PWM, GPIO, Hardware Timers, Interrupts
- **Frameworks & Core Skills:** FreeRTOS, Bare-Metal Programming, Hardware Debugging, Datasheet Analysis, IoT, Signal Processing
- **Tools & EDA:** Synopsys VCS, TCAD, Cadence Virtuoso, ESP-IDF, Xilinx Vivado, LabVIEW, Tinkercad, Git/GitHub.

## PROJECTS

**Context-Aware Hazard Monitor & Voice Alert System** | **Presentation** — *Arduino IDE, Embedded C, ADC, DAC*

- Engineered an IoT environmental hazard monitor using Embedded C via the Arduino IDE, analyzing technical datasheets and configuring hardware ADC channels to calibrate MQ-2 gas, DHT11, and LDR sensors.
- Developed a weighted multi-parameter risk-scoring algorithm that processes sensor data to classify environmental conditions across 4 severity levels.
- Implemented an offline DAC-driven voice alert subsystem streaming raw 8-bit PCM audio from PROGMEM to the internal DAC and PAM8403 amplifier, delivering speech alerts independently.
- Configured secure Firebase communication rules via WiFi libraries to pair each device to a single user.

**Real-Time Morse Code Detection & Display System** | **Presentation** — *ESP-IDF, FreeRTOS, Hardware Timers, GPIO*

- Engineered a real-time Morse code decoder using the ESP-IDF framework, achieving 99% signal classification accuracy through custom hardware timers and GPIO interrupts.
- Implemented FreeRTOS tasks, queues, and ISRs for efficient, non-blocking pulse-width timing evaluation (250ms dot/dash discrimination and 600ms gaps) ensuring zero data loss.
- Interfaced a standard 16-pin (16x2) parallel LCD using bare-metal GPIO manipulation for instant text visualization alongside synchronous piezo buzzer audio.

**AQMD System & Peripheral Simulations** | **Presentation** — *Completed | IoT, UART/I2C, PWM*

- Architected a portable smart water testing device providing real-time assessments of pH, ammonia, and nitrate levels, linking distinct environmental sensors to a continuous 24/7 dashboard.
- Formulated an automated sensor cleaning mechanism using simulated PWM motor controls via Tinkercad, reducing manual maintenance and cutting farmer operational costs.
- Simulated and validated device-to-device communication using UART and I2C protocols during the prototyping phase to ensure robust data transfer.

## EXPERIENCE

**Embedded Systems Intern** *[Offer]* — 10/2025 – 03/2026
*SURE Trust / ProEd* — *Remote*

- Engineered firmware in Embedded C for IoT applications, including a context-aware text-to-speech hazard monitor utilizing an ESP32, FreeRTOS, and hardware ADCs for real-time multi-sensor data acquisition.
- Interfaced environmental sensors and hardware peripherals using UART, I2C, and GPIO interrupts, while validating communication protocols and system reliability for continuous 24/7 dashboards.

**Summer Research Intern (VLSI Signal Processing)** *[Post]* — 05/2026 – 07/2026
*IIT Bhubaneswar (Dept. of ECE)* — *Bhubaneswar, Odisha*

- Entrusted with full ownership of high-performance BCI signal processing pipelines across 4 massive datasets (>50GB); implemented Butterworth bandpass filtering and Euclidean Alignment (EA) for spatial covariance normalization.
- Applied Riemannian Tangent Space projection and Hybrid Stacking Ensembles (SVM, LDA, RF) to optimize EEG classification frameworks.

**Industrial Trainee (RF & Semiconductors)** *[Certificate]* — 05/2026 – 06/2026
*GAETEC (DRDO / ISRO Foundry)* — *Hyderabad, TS*

- Analyzed MESFET and pHEMT device architectures in the Design section using Keysight tools, focusing on Gallium Arsenide (GaAs) fabrication across high-frequency RF spectrums.
- Evaluated ATQC (Assembly, Testing & QC) processes for 100+ components: Die/Wire bonding, Shear/Pull testing, and RF characterization via VNA/PNA; assessed PIND, Acceleration, and Thermal cycles.

## KEY ACHIEVEMENTS

- **1st Runner-Up (Spark Tank):** Secured an INR 30k cash prize among 50+ teams from E-Cell VITB for the AQMD project.
- **1st Place (Hustlethon 2k25):** Engineered the winning IoT Aquaculture Water Testing solution out of 100+ competitors.
- **Advanced RISC-V Tapeout:** Completed an intensive 10-week RTL-to-GDSII flow for a 32-bit RISC-V core, ranking top 22 nationally. *[View GitHub Repo]*
- **HAM Radio Operator:** Hold a General License; actively operate at the Vishnu Space Applications Center.
- **Certifications:** HCL VLSI Foundation | NPTEL Elite Comm. Systems | VLSI SoC Design | NPTEL Elite+ IoT